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Highly Efficient Dissolution of Lignin by Eutectic Molecular Liquids

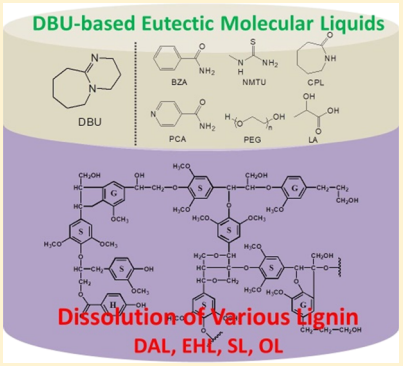
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Supporting Information

ABSTRACT: Lignin dissolution is of great importance for the valorization of lignin, and development of novel efficient solvent systems is a crucial topic. Herein, 17 kinds of 1,5-diazabicyclo[5.4.0]-5-undecene-based eutectic molecular liquids (EMLs) were designed for dissolution of industrial lignins. The solubility of lignin in the EMLs significantly depended on the types of the EMLs and lignin. Also the relationship between lignin solubility and the solvatochromic parameters (α, β, and π^*) varied significantly based on the EMLs and the types of lignin.



INTRODUCTION

As the only renewable natural aromatic biopolymer, lignin could be utilized to produce various aromatic chemicals and functional materials,[1−5] which has received continuously increasing attention. Generally, an adequate solvent is necessary in most of the processes for lignin utilization. However, lignin was difficult to be dissolved in most common solvents owing to its complicated cross-linked three-dimensional structure.[6,7] Therefore, development of functional solvents to dissolve lignin is of great significance for lignin utilization, and much effort has been devoted on this important topic.

To date, several solvent systems have been developed for lignin dissolution, but some specific drawbacks existed in the corresponding systems. For example, the mixture of water and some organic solvents (e.g., polysorbate,[8] γ-valerolactone,[9,10] and ethanol[11,12]) could be employed to dissolve lignin, but the high volatility of organic solvents was the main drawback of these systems. As the alternatives of organic solvent-based systems, ionic liquid (IL)-based systems have been successfully applied in lignin dissolution.[13−17] In spite of the good solubility of lignin, IL-based systems were limited by the high cost, not so good stability, and the use of toxic reagents and/or organic solvents in their synthetic routes.[18,19] Additionally, liquid ammonia had the ability to dissolve lignin,[20] but high pressure and too low temperature hampered its large-scale application. In another aspect, it is well-known that the resources and preparation methods significantly affected the structures and properties of lignin[21−23] and thus influenced the solubility of different lignin in one specific solvent. Based on the above discussion, it is a highly attractive issue to design functional solvents with the purpose of efficiently and applicably dissolving different types of lignin.

Deep eutectic solvents (DESs) have attracted tremendous attention as a class of green functional solvents applied in diverse areas.[24−28] Recently, several DESs has been successfully used in the dissolution of lignin.[29−31] However, the current developed DESs were generally based on quaternary ammonium salts, and their high viscosity limited their large-scale and practical applications for lignin dissolution due to the mass transfer between lignin and DESs.[32,33]

Recently, eutectic molecular liquids (EMLs), an emerging type of solvents, have been proposed.[34,35] EMLs can be generated from two or more molecular compounds via various noncovalent interactions (e.g., hydrogen bonding, σ-hole bonding, π-hole bonding, κ-hole bonding, and μ-hole bonding).[34] Compared with ionic DESs, EMLs could be formed from low-cost materials with lower viscosity. Meanwhile, the components in EMLs could be easily recovered and regenerated by evaporation, a significant merit over the general ionic DESs.[36] Additionally, similar with ionic DESs, the properties of EMLs could be easily tuned by changing the components. These advantages make EMLs an attractive type of solvents for lignin dissolution, and novel solvents based on EMLs could be potentially designed to dissolve lignin efficiently.

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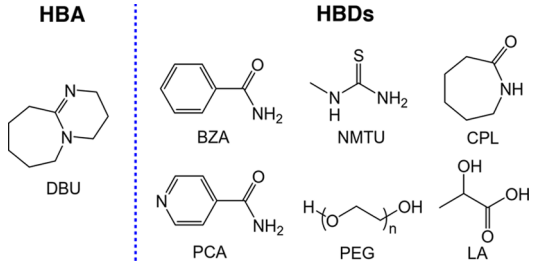
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It has been reported that 1,5-diazabicyclo[5.4.0]-5-undecene (DBU) could promote the dissolution of cellulose in dimethyl sulfoxide (DMSO) in the presence of CO_2.[37] Additionally, DBU could be employed as a hydrogen bond acceptor to interact with methylthiourea to dissolve V_2VI_3 chalcogenides.[38] These works gave us a cue that DBU might be used to form EMLs with great potential as good solvents for lignin dissolution.

Herein, 17 kinds of DBU-based EMLs (denoted as DBU-HBD-*x*, *x* represented the molar ratio of DBU and hydrogen bond donor (HBD)) were designed and synthesized by employing DBU and six HBDs as described in Scheme 1 (i.e., *ε*-caprolactam (CPL), benzamide (BZA), 4-pyridine carboxamide (PCA), polyethylene glycol 200 (PEG), lactic acid (LA) and N-methylthiourea (NMTU)). By carefully pairing DBU with a suitable HBD and modifying the molar ratio of DBU/HBD, these DBU-based EMLs could efficiently dissolve different types of lignin, including enzymatic hydrolysis lignin (EHL), organosolv lignin (OL), dealkaline lignin (DAL), and sodium lignosulfonate (SL).

Scheme 1. Chemical Structures of DBU and the Used HBDs



RESULTS AND DISCUSSION

Synthesis of DBU-Based EMLs. The DBU-based EMLs were initially synthesized by heating the mixture of DBU and the desired HBD (Scheme 1) with a suitable molar ratio. The detailed synthetic routes were provided in the Experimental Section. Meanwhile, the formed temperature for a specific EML and the suitable molar ratio of DBU/HBD capable of forming EMLs were provided in Table S1. It was revealed that a higher molar ratio of DBU/HBD was beneficial for the formation of EMLs at a lower heating temperature. For example, EMLs could be formed between DBU and all of the six HBDs with a DBU/HBD molar ratio of 2:1, and only PCA could not form EML with DBU at a DBU/HBD molar ratio of 1:1. However, when the DBU/HBD molar ratio was changed into 1:2 or 1:3, only three used HBDs (i.e., CPL, LA, and PEG) could generate EMLs with DBU. In addition, for practical applications, the physicochemical properties (i.e., viscosity and density) of the achieved EMLs were determined and provided in Tables S2 and S3.

Dissolution Behavior of Lignin in the DBU-Based EMLs. Dealkaline lignin (DAL) was selected as a model substrate to evaluate the ability of the synthesized EMLs for lignin dissolution. It was observed that almost all the prepared EMLs provided good solubility except for DBU-PEG-2:1 (Figure 1 and Table S4), and the solubility of DAL increased with the rising of temperature (from 30 to 90 °C) for all the formed EMLs because the viscosity of the EMLs decreased with the increased temperature (Table S2),[39,40] which was beneficial for DAL dissolution. When the DBU/HBD molar ratio was 2:1, the EMLs generated using LA, CPL, BZA, and PCA showed good performance for DAL dissolution (Figure 1a), and the highest solubility of 39.6 wt% could be achieved at 90 °C in DBU-CPL-2:1 (Table 1, entry 1). When the DBU/HBD molar ratio was changed to 1:1, the EMLs from LA, PEG, and CPL provided the satisfactory solubilities of DAL, and DBU-CPL-1:1 and DBU-PEG-1:1 showed similar results (Figure 1b) with the highest value of 41.7 and 40.9 wt% (Table 1, entries 2 and 3) at 90 °C, respectively. Furthermore, when the DBU/HBD molar ratio was decreased to 1:2 or 1:3, LA and PEG became the preferred HBDs (Figure 1c). The highest solubility was 48.2 wt% in DBU-LA-1:2 at 90 °C (Table 1, entry 4) at the DBU/HBD molar ratio of 1:2, while the best

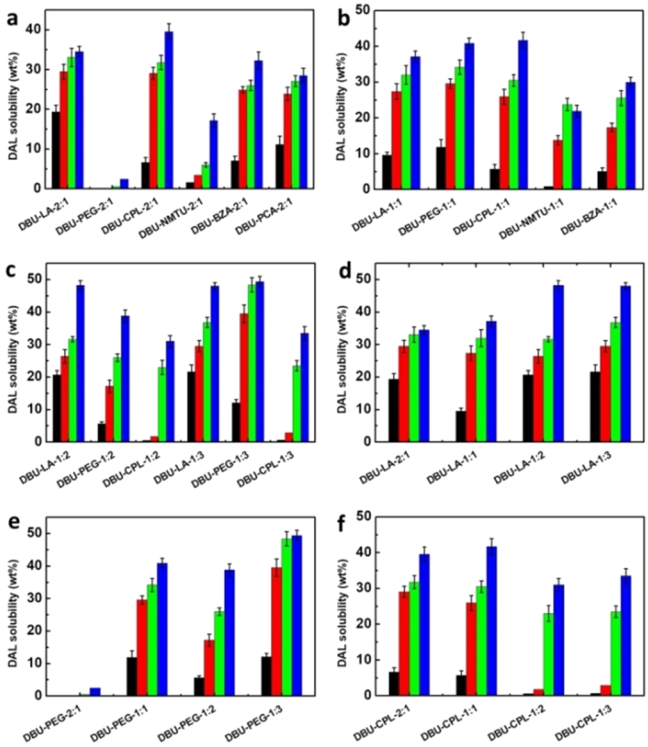


Figure 1. Solubilities of DAL in the synthesized EMLs at different temperatures. The dark, red, green, and blue bars represent the experiments conducted at 30, 50, 70, and 90 °C, respectively.

Table 1. Best Results for Different Types of Lignin in the Prepared EMLs at 90 °C

entry	lignin	EML	solubility (wt%)[a]
1	DAL	DBU-CPL-2:1	39.6 (±2.0)
2	DAL	DBU-CPL-1:1	41.7 (±2.3)
3	DAL	DBU-PEG-1:1	40.9 (±1.5)
4	DAL	DBU-LA-1:2	48.2 (±1.5)
5	DAL	DBU-PEG-1:3	49.4 (±1.6)
6	EHL	DBU-NMTU-2:1	32.7 (±2.4)
7	EHL	DBU-PEG-1:1	38.8 (±1.6)
8	EHL	DBU-PEG-1:2	42.1 (±2.2)
9	EHL	DBU-PEG-1:3	41.1 (±1.0)
10	OL	DBU-LA-1:3	37.3 (±1.6)
11	SL	DBU-LA-1:3	38.3 (±1.9)

[a]Values of solubility were the average of twice measurements, and the values in the parentheses represented for the half of the range of replicate results

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Ind. Eng. Chem. Res. 2019, 58, 23438−23444

choice changed to DBU-PEG-1:3 with a highest solubility of 49.4 wt% at 90 °C (Table 1, entry 5) when the DBU/HBD molar ratio became 1:3. For the EMLs synthesized from the same HBD, the types of EML affected the dissolution behavior significantly (Figure 1d–f). When LA was used as the HBD, DBU-LA-1:2 and DBU-LA-1:3 provided a similar solubility, which were better than DBU-LA-2:1 and DBU-LA-1:1 (Figure 1d). In contrast, the solubility increased with the decrease of the DBU/PEG molar ratio in the EMLs using PEG as the HBD (Figure 1e), and DBU-PEG-1:3 was the best EML among the four DBU-PEG-*x*. However, when the HBD was CPL, DBU-CPL-2:1 and DBU-CPL-1:1 were the suitable EMLs (Figure 1f). In one word, the influence of DBU/HBD molar ratio on DAL solubility varied with the types of HBDs, and the solubility of DAL in EMLs with the specific HBD followed the orders: DBU-LA-1:3 > DBU-LA-1:2 ≈ DBU-LA-2:1 > DBU-LA-1:1, DBU-PEG-1:3 > DBU-PEG-1:1 > DBU-PEG-1:2 > DBU-PEG-2:1, and DBU-CPL-2:1 > DBU-CPL-1:1 > DBU-CPL-1:2 ≈ DBU-CPL-1:3. In addition, the EMLs using LA and PEG as the HBDs showed better performance than those with amides (BZA, PCA, and CPL) and NMTU as the HBDs in most cases to dissolve DAL.

The dissolution of the other three types of lignin (i.e., EHL, OL, and SL) in the prepared EMLs was subsequently determined. First, for EHL dissolution, DBU-PEG-2:1, DBU-CPL-2:1, DBU-CPL-1:1, DBU-CPL-1:2, and DBU-CPL-1:3 showed very poor ability (Table S5), and the performance of other EMLs was then examined for EHL dissolution. It was discovered that the solubility of EHL in these EMLs increased with the increasing temperature (Figure 2), owing to the

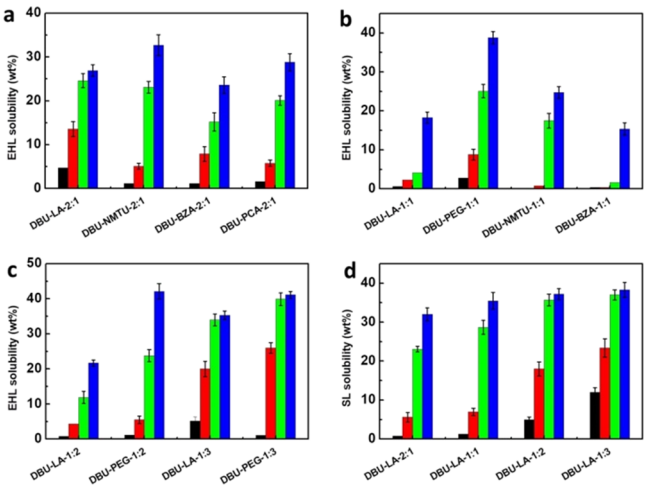


Figure 2. (a–c) Solubilities of EHL in the synthesized EMLs and (d) SL in DBU-LA-*x* at different temperatures. The dark, red, green, and blue bars represent the experiments conducted at 30, 50, 70, and 90 °C, respectively.

decreased viscosity. With a DBU/HBD molar ratio of 2:1, the ability of the corresponding EMLs to dissolve EHL was similar (Figure 2a), and the highest solubility was 32.7 wt% at 90 °C in DBU-NMTU-2:1 (Table 1, entry 6). When the DBU/HBD molar ratio became 1:1 (Figure 2b), DBU-PEG-1:1 was the best EML with a 38.8 wt% solubility at 90 °C (Table 1, entry 7). However, when the DBU/HBD molar ratio continually decreased to 1:2 and 1:3, only the EMLs formed with LA and PEG could dissolve EHL. Meanwhile, DBU-PEG-*x* showed better performance than DBU-LA-*x* at the same DBU/HBD molar ratio (Figure 2c), and the solubilities for DBU-PEG-1:2 and DBU-PEG-1:3 were 42.1 and 41.1 wt% at 90 °C (Table 1, entries 8 and 9). In another point of view, for the EMLs with the same HBD, the influence of the DBU/HBD molar ratio on EHL solubility varied significantly with the HBD types, and the orders were as follows: DBU-LA-1:3 > DBU-LA-2:1 > DBU-LA-1:2 > DBU-LA-1:1, DBU-BZA-2:1 > DBU-BZA-1:1, DBU-NMTU-2:1 > DBU-NMTU-1:1, and DBU-PEG-1:3 > DBU-PEG-1:2 ≈ DBU-PEG-1:1 > DBU-PEG-2:1. Second, as shown in Table S6, OL was difficult to be dissolved in most of the synthesized EMLs, and only DBU-LA-*x* could dissolve OL in all the used temperatures in most cases. The solubility of OL in DBU-LA-*x* followed the order: DBU-LA-1:3 > DBU-LA-1:2 > DBU-LA-2:1 > DBU-LA-1:1, and the highest solubility of 37.3 wt% was achieved in DBU-LA-1:3 at 90 °C (Table 1, entry 10). In addition, several EMLs could provide the satisfactory solubility of OL at a higher temperature (90 °C), including DBU-NMTU-2:1 (19.7 wt%), DBU-BZA-2:1 (12.7 wt%), DBU-PCA-2:1 (30.6 wt%), DBU-NMTU-1:1 (18.7 wt%), DBU-BZA-1:1 (22.1 wt%), DBU-PEG-1:2 (23.8 wt%), and DBU-PEG-1:3 (11.4 wt%). Third, it was found that only DBU-LA-*x* could dissolve SL (Table S7), which may be resulted from its salt-like structure. Meanwhile, the solubility of SL in DBU-LA-*x* increased with the decreasing DBU/LA molar ratio (Figure 2d), and DBU-LA-1:3 provided the highest solubility (38.3 wt%) at 90 °C (Table 1, entry 11). For comparison, the solubility of lignin from other literatures was listed in Table S8. It can be seen that the reported solvents either have poor solubility for lignin (e.g., the solubility of alkali lignin in malic acid/proline (1:3) is limited to 14.9 wt% even at 100 °C) or are expensive and environmentally non-benign (e.g., $[P_{4441}]$-$[MeSO_4]$, $[BMPyr][N(CN)_2]$).

On the basis of the above experiments, we could find that the solubility of lignin significantly depended on the types of the DBU-based EMLs (type of HBDs and DBU/HBD molar ratio) and lignin. It was observed that DBU-LA-*x* could dissolve all the four types of lignin at all examined temperatures except that DBU-LA-1:1 only showed good performance for OL dissolution at high temperatures (70 and 90 °C), indicating DBU-LA-*x* was the most adaptable EML to fit different types of lignin. Meanwhile, DAL could be dissolved by all the obtained EMLs except for DBU-PEG-2:1, while the types of EMLs, which could dissolve EHL, were less than those for DAL. Furthermore, OL was more difficult to be dissolved, and only DBU-LA-*x* could dissolve OL at all examined temperatures in most cases, while a high temperature (90 °C) was necessary for several other EMLs to dissolve OL (Table S6). Additionally, only DBU-LA-*x* could dissolve SL, and all other EMLs showed very poor solubility for SL even at 90 °C. These results implied that DAL was the most easily dissolved lignin because of the serious destruction of β-O-4 linkage in the preparation process, which was helpful for DAL dissolution, and the difficulty to dissolve SL was resulted from its salt-like structure.

Mechanism on the Different Dissolution Ability of the DBU-Based EMLs. The ability to dissolve lignin in one solvent might correlate with the Kamlet–Taft empirical parameters (i.e., α, β, and π^*) of the solvent,[9] which quantify the hydrogen-bond donating ability, hydrogen-bond accepting ability, and polarity of the solution.[41] Herein, with the purpose to provide some deep information about the ability of the synthesized EMLs to dissolve different types of lignin, the Kamlet–Taft empirical parameters were determined (Table

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Ind. Eng. Chem. Res. 2019, 58, 23438–23444

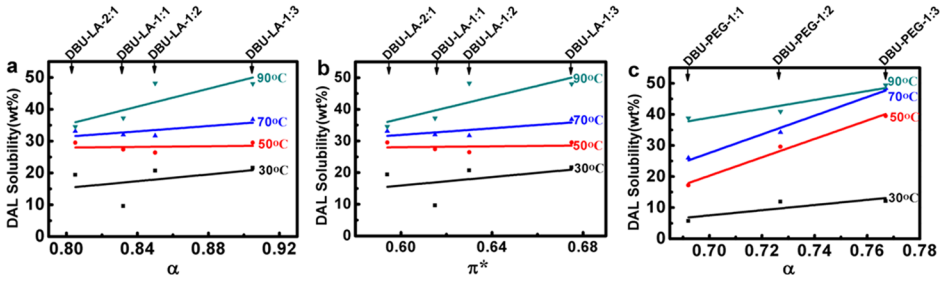


Figure 3. Correlation between the DAL solubility and Kamlet−Taft empirical parameters. (a, b) DAL dissolved in DBU-LA-*x* and (c) DAL dissolved in DBU-PEG-*x*.

S9) based on some reported methods.[42,43] At first, a simple linear correlation of the solubility of the lignins with the Kamlet−Taft parameters of the EMLs were performed, and the results are given in Figures 3−5. For a reasonable comparison,

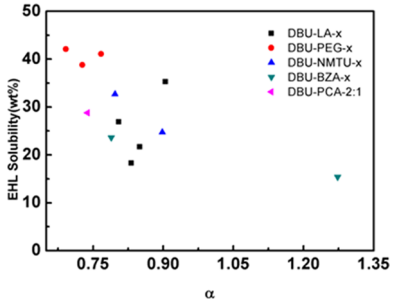


Figure 4. Correlation between the EHL solubility and Kamlet−Taft empirical parameters of all EMLs.

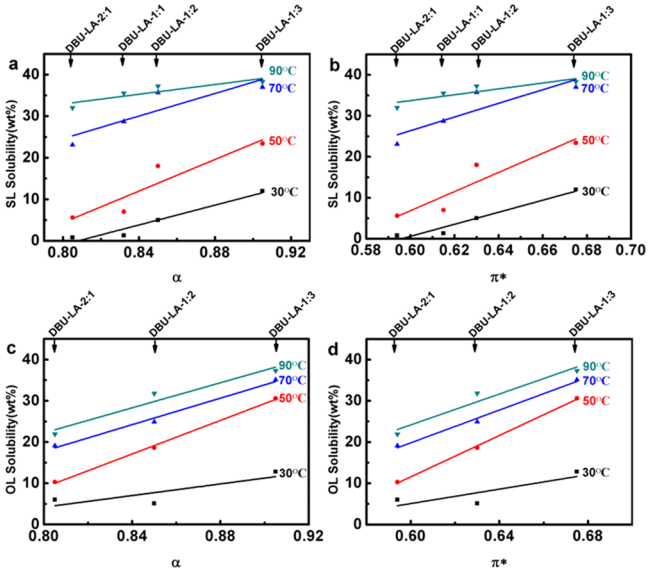


Figure 5. Correlation between the solubility of (a) SL and (b) OL and the Kamlet−Taft empirical parameters of DBU-LA-*x*.

we only discussed the relationship between lignin solubility and the Kamlet−Taft empirical parameters of the EMLs with the same HBD. Furthermore, a Kamlet−Taft linear solvation energy relationship (LSER) for quantifying solvent effects, which were introduced to investigate the correlations between the solubility and the Kamlet−Taft parameters of corresponding EMLs, were carried out, and the results are presented in Table 2.[44,45]

Table 2 shows that in most cases, the solubility of lignin is shown to be related to the hydrogen-bond donor property (α) and the Kamlet−Taft π^* parameter of corresponding EMLs.

First, from Figure 1d, the four DBU-LA-*x* could dissolve DAL efficiently with a solubility order of DBU-LA-1:3 > DBU-LA-1:2 ≈ DBU-LA-2:1 > DBU-LA-1:1. Table 2 and Figure 3b shows that there are positive correlations between the DAL solubilities and the Kamlet−Taft empirical parameters α and π^*, but the correlation with β is not good. However, if the DBU-LA-2:1 is excluded, the positive correlations (DAL solubility with α and π^*) and a negative correlation (DAL solubility with β) could be observed (Figure S1). The correlation between DAL solubility at 70 °C with the Kamlet−Taft empirical parameters follows the equation $\ln S = 1.09 + 4.11\pi^*$ (Table 2). Also, DBU-LA-1:3 was the best EML in the DBU-LA-*x* due to its highest polarity (π^*) ability. Meanwhile, the three DBU-PEG-*x* could dissolve DAL with a dissolution capability of DBU-PEG-1:3 > DBU-PEG-1:1 > DBU-PEG-1:2 (DBU-PEG-2:1 was not considered because DAL is almost insoluble in DBU-PEG-2:1). Table S9 shows that the values of α, β, and π^* changed as the following orders: DBU-PEG-1:3 > DBU-PEG-1:2 > DBU-PEG-1:1 for π^*, DBU-PEG-1:3 > DBU-PEG-1:1 > DBU-PEG-1:2 for α, and DBU-PEG-1:3 > DBU-PEG-1:1 > DBU-PEG-1:2 for β. The correlation between DAL solubility at 70 °C with the Kamlet−Taft empirical parameters obeys the equation: $\ln S = -2.49 + 8.29\alpha$ (Table 2 and Figure 3c). DBU-PEG-1:3 showed the highest solubility for DAL owing to its strongest hydrogen-bond donating ability (the highest value of α) to form hydrogen bonding with DAL. Furthermore, for the four DBU-CPL-*x* (the order for DAL solubility: DBU-CPL-2:1 > DBU-CPL-1:1 > DBU-CPL-1:2 ≈ DBU-CPL-1:3), no obvious correlations were observed between the DAL solubility and the Kamlet−Taft empirical parameters (i.e., α, β, π^*, and even α-β, Table S10). Additionally, for DBU-NMTU-*x*, the DAL solubility increased with the values of α, β, and π^* of DBU-NMTU-*x* (Table S11). However, for DBU-BZA-*x*, the solubility of DAL at 70 °C in DBU-BZA negatively correlated with the values of both α and π^* (Table 2 and Table S12).

Second, EHL could be dissolved in all the four DBU-LA-*x* with a dissolution capability of DBU-LA-1:3 > DBU-LA-2:1 > DBU-LA-1:2 > DBU-LA-1:1. As shown in Table S9, the values of α, β, and π^* changed as the following orders: DBU-LA-1:3 > DBU-LA-1:2 > DBU-LA-1:1 > DBU-LA-2:1 for α and π^* and DBU-LA-1:1 > DBU-LA-1:2 > DBU-LA-1:3 > DBU-LA-2:1 for β. It was obvious that there was no correlation between the dissolution capability of DBU-LA-*x* and the three Kamlet−Taft empirical parameters. Meanwhile, DBU-PEG-*x* could also dissolve EHL with a dissolution capability of DBU-PEG-1:3 > DBU-PEG-1:2 ≈ DBU-PEG-1:1 > DBU-PEG-2:1. Unfortu-

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Table 2. Results of Kamlet−Taft LSER Fits for the Dissolution of Lignin in Different Kinds of Eutectic Molecular Liquids

lignin	EMLs	parameters	LSER equation	*p*-values
EHL	all kinds of EMLs	α (90 °C)	$\ln S = 4.58-1.49\alpha$	$XYZ_0 = 6 \times 10^{-7}$
				$\alpha = 0.01$
EHL	DBU-NMTU	α (90 °C)	$\ln S = 4.26-1.20\alpha$	$XYZ_0 = 7 \times 10^{-4}$
	DBU-BZA			$\alpha = 0.03$
	DBU-PCA			
DAL	DBU-LA	π^* (90 °C)	$\ln S = 1.09 + 4.11\pi^*$	$XYZ_0 = 0.05$
				$\pi^* = 0.02$
DAL	DBU-PEG	α (70 °C)	$\ln S = -2.49 + 8.29\alpha$	$XYZ_0 = 0.05$
				$\alpha = 0.02$
DAL	DBU-NMTU	α, π^* (70 °C)	$\ln S = 3.86-0.02\alpha - 1.00\pi^*$	$XYZ_0 = 7 \times 10^{-4}$
	DBU-BZA			$\alpha = 0.04$
	DBU-PCA			$\pi^* = 0.004$
OL	DBU-LA	α (70 °C)	$\ln S = -1.96 + 6.09\alpha$	$XYZ_0 = 0.03$
				$\alpha = 0.01$
OL	DBU-LA	π^* (70 °C)	$\ln S = -1.52 + 7.52\pi^*$	$XYZ_0 = 0.02$
				$\pi^* = 0.006$

nately, there were no obvious correlations between the EHL solubility and the Kamlet−Taft empirical parameters (i.e., α, β, and π^*, Table S13). The only conclusion we could obtain was that DBU-PEG-1:3 was the best choice in DBU-PEG-*x* for EHL dissolution. Moreover, for the DBU-BZA-*x*, the solubility of EHL correlated positively with the β value but had a negative correlation with the values of both α and π^* (Table S14). The EHL solubility of all EMLs at 90 °C decreased with the increased ability of hydrogen-bond donating (α), which follows the formula $\ln S = 4.58-1.49\alpha$.

Third, only DBU-LA-*x* showed better performance for the dissolution of SL and OL, and the solubilities of SL and OL in DBU-LA-*x* followed the order: DBU-LA-1:3 > DBU-LA-1:2 > DBU-LA-1:1 > DBU-LA-2:1 for SL and DBU-LA-1:3 > DBU-LA-1:2 > DBU-LA-2:1 > DBU-LA-1:1 for OL. For SL, positive correlations were found between the solubility of SL and the values of α and π^* (Figure 5a,b), but a negative correlation between SL solubility and the β value would be observed if DBU-LA-2:1 was excluded (Figure S2). Meanwhile, for OL, DBU-LA-1:1 was not considered, owing to the very low solubility of OL. For other three DBU-LA-*x*, positive correlations were also found between OL solubility and the values of α and π^* (Figure 5c,d), but no correlation existed between OL solubility and the β value (Figure S3). The above results indicated that a higher polarity (π^*) and the hydrogen-bond donating ability (α) were beneficial for the dissolution of SL and OL, and thus DBU-LA-1:3 showed the highest solubilities for OL, owing to its highest values of α and π^*.

As discussed above, the relationship between the solubility of lignin and the values of solvatochromic parameters (i.e., α, β and π^*) is complicated, which is attributed to the complex structure of lignin. For example, wood can be degraded by an acid and alkali solution in the pulping and papermaking process, rather than an acid solution or alkali solution, while cellulose or chitosan only linearly related to the β or $\beta-\alpha$ parameter.[46,47] The relationship between the lignin solubility and the values of solvatochromic parameters significantly varied from the type of lignin when the used EMLs possessed a specific HBD. As shown in Tables S4−S7 and S9, if all of the achieved EMLs were compared together, the correlation between lignin solubility and the values of solvatochromic parameters was much worse, indicating that the nature of the used EMLs may be the key parameter to affect lignin dissolution. Therefore, it could be concluded that the nature of the synthesized EMLs and the types of lignin were crucial for the relationship between lignin solubility and the solvatochromic parameters. When the ability of EMLs to dissolve lignin was estimated, this point must be paid more attention, and it would be no reason if only the solvatochromic parameters were compared.

CONCLUSIONS

In conclusion, 17 kinds of DBU-based EMLs were designed using six HBDs (i.e., CPL, BZA, PCA, PEG, LA, and NMTU) with the goal to dissolve lignin efficiently. It was found that the solubility significantly depended on the nature of the EMLs and the types of lignin. DAL could be dissolved in all the obtained DBU-based EMLs except for DBU-PEG-2:1, and most of the EMLs could dissolve EHL. However, OL was more difficult to be dissolved than DAL and EHL. Only DBU-LA-*x* (except for DBU-LA-1:1 with very low solubility) could dissolve OL at all examined temperatures, and higher temperatures (70 and 90 °C) were needed for other types of EMLs to dissolve OL. Meanwhile, DBU-LA-*x* was the only type of EMLs to dissolve SL for all examined temperatures. The systematic study revealed that the nature of the synthesized EMLs and the types of lignin significantly affected the relationship between lignin solubility and the solvatochromic parameters. Additionally, the DBU-based EMLs using LA and PEG as the HBDs showed better performance than those with amides (BZA, PCA, and CPL) and NMTU as the HBDs in most cases. We believe that the DBU-based EMLs have great potential to be utilized as efficient task-specific solvents for lignin dissolution, and more EMLs could be hopefully designed for efficient dissolution of lignin, which are ongoing in our group.

EXPERIMENTAL SECTION

Materials. Benzamide (BZA) and 1,8-diazabicyclo[5.4.0]-undec-7-ene (DBU) were obtained from Beijing Innochem Technology Co., Ltd. Polyethylene glycol 200 (PEG) was purchased from China Pharmaceutical Group Chemical Reagents Co., Ltd. Lactic acid (LA) was obtained from McLean Biochemical Technology Co., Ltd. (Shandong, China). Pyridine-4-carboxylic acid amide (PCA) and N-methylthiourea (NMTU) were purchased from Aladdin

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Ind. Eng. Chem. Res. 2019, 58, 23438−23444

Biochemical Technology Co., Ltd. (Shandong, China). ε-Caprolactam (CPL) was supplied by Alfa Aesar (China) Chemical Co., Ltd. Dealkaline lignin (DAL) was purchased from Sigma Aldrich (Shanghai) Trading Co., Ltd. The enzymatically hydrolyzed lignin (EHL) was isolated from the cellulolytic enzyme hydrolysis of corn cob and supplied by Shandong Longlive Bio-Technology Co., Ltd. (Shandong, China).[48] Formic acid organosolv lignin (OL) and sodium lignosulfonate (SL) were prepared from dewaxed bamboo (Yunnan, China).[49] Nile red and 4-nitroaniline were achieved from J&K Scientific.

Preparation and Characterization of EMLs. In a typical route, DBU and the suitable HBDs (i.e., CPL, BZA, PCA, PEG, LA, and NMTU) were mixed at the molar ratio of 1:1, 1:2, 1:3, or 2:1. Then, the corresponding mixture was heated on a magnetic stirrer with a heating temperature of 25 to 60 °C (Please see Table S1) until a homogeneous liquid was generated. At last, the achieved liquids were heated at 60 °C for 24 h under vacuum to remove the contained water, and the desired EMLs were finally obtained.

The examination of the viscosity of the synthesized EMLs was conducted on an Anton Paar AMVn viscosity meter using a falling ball technique. Double distilled water was used to calibrate the viscosity meter, and the repeatability uncertainty was controlled at 0.01. Meanwhile, the density of the prepared EMLs was measured using an Anton Paar DMA 5000 M density meter with the accuracy of $\pm 5 \times 10^{-6}$ g cm^{-3}. The temperatures for the detection of viscosity and density were 25, 35, 45, and 55 °C.

Solubility of Lignin. The solubility of lignin in the prepared EMLs was determined through a two-step process.[31] Since it is the same with the previous reported method, the detailed procedure for the determination is only provided in the Supporting Information. It should be pointed out that the obtained solubility value was the average of two measurements, and the errors for the solubility were the half of the range of replicate results in the corresponding figures (Figures 1−5 and Figures S1−S3) and tables (Table 1 and Tables S4−S7 and S9−S13).

Solvatochromic Parameters Measurement. The determination of the solvatochromic parameters (α, β and π^*) had been described in many papers.[9,31] It was determined by measure of the maximum absorption wavelength of two dyes, Nile red and 4-nitroaniline. A brief description of the procedure of the determination is given in the Supporting Information.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.iecr.9b05059.

Experimental sections, solubility determination and solvatochromic parameters measurement; the viscosity (η), density (ρ), and solvatochromic parameters of the prepared EMLs; the temperature for the formation of the EMLs, solubility of various lignins at different temperatures; the relationship between the lignin solubility in the prepared EMLs; and the references (PDF)

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Notes

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